WHATSNEXT VISION MOTORS

Salesforce CRM Implementation

Vehicle Ordering • Dealer Assignment • Stock Validation • Test Drive Automation

Project	WhatsNext Vision Motors CRM
Platform	Salesforce Developer Org
Core Technologies	Custom Objects • Flow • Apex • Batch Apex • Scheduled Apex
Primary Goal	Automate vehicle ordering and improve customer experience
Document Type	Model Project / Implementation Guide

Prepared as a clean reference model from the supplied SkillWallet project instructions and screenshots.

1. Project Overview

WhatsNext Vision Motors is implementing Salesforce CRM to streamline vehicle ordering, dealer operations, stock management, and test-drive communication.

The solution uses Salesforce configuration plus Apex automation. Orders are validated against available vehicle stock, confirmed orders reduce stock, pending orders can be processed automatically when stock becomes available, and scheduled reminders are sent for upcoming test drives.

The implementation is designed to improve order accuracy, reduce manual work, and provide a consistent customer experience.

Key business capabilities:

- Maintain Vehicle, Vehicle Dealer, Customer, Vehicle Order and Vehicle Test Drive information.
- Provide a dedicated Salesforce Lightning application named **WhatNext Vision Motors**.
- Prevent orders when the selected vehicle has no available stock.
- Reduce vehicle stock when an order is confirmed.
- Automatically process pending orders when new stock becomes available.
- Send an email reminder before a scheduled test drive.
- Run batch order processing on a recurring schedule.

High-level automation flow

Customer	Vehicle Order	Stock Validation	Confirmation	Stock Update
Places order	Created/Updated	Apex Trigger	Confirmed/Pending	Vehicle quantity adjusted

2. Creating the Salesforce Developer Account

Create a Salesforce Developer Org before starting the configuration.

Signup URL: https://developer.salesforce.com/signup

Enter the following details on the signup form:

- **First Name / Last Name:** Enter your name.
- **Email:** Use an accessible email address.
- **Role:** Developer.
- **Company:** College name.
- **Country:** India.
- **Postal Code:** Your PIN code.
- **Username:** Use a unique username in Salesforce username format. It does not have to be a real email inbox.

Click Sign Me Up. Open the Salesforce verification email, verify the account, set the password/security details, and then log in to the Developer Org.

Reference screenshot

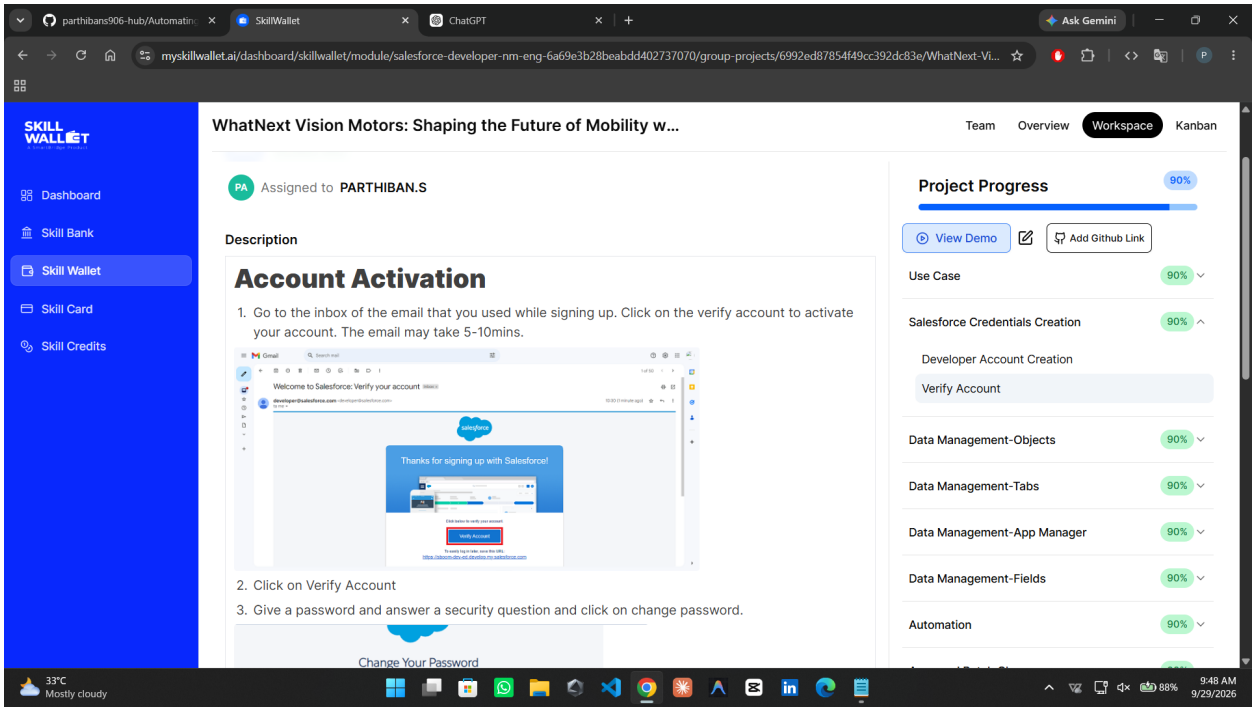


Figure 1. SkillWallet project page showing Salesforce account activation instructions.

3. Creating the Vehicle Dealer Custom Object

The Dealer object stores detailed information about dealers participating in the vehicle ordering process.

Navigation: Setup → Object Manager → Create → Custom Object

Field	Value
Label	Vehicle Dealer
Plural Label	Vehicle Dealers
Record Name Label	Dealer Name
Data Type	Text
Additional Options	Allow Reports; Allow Search

Click **Save** after completing the object definition.

Reference screenshot

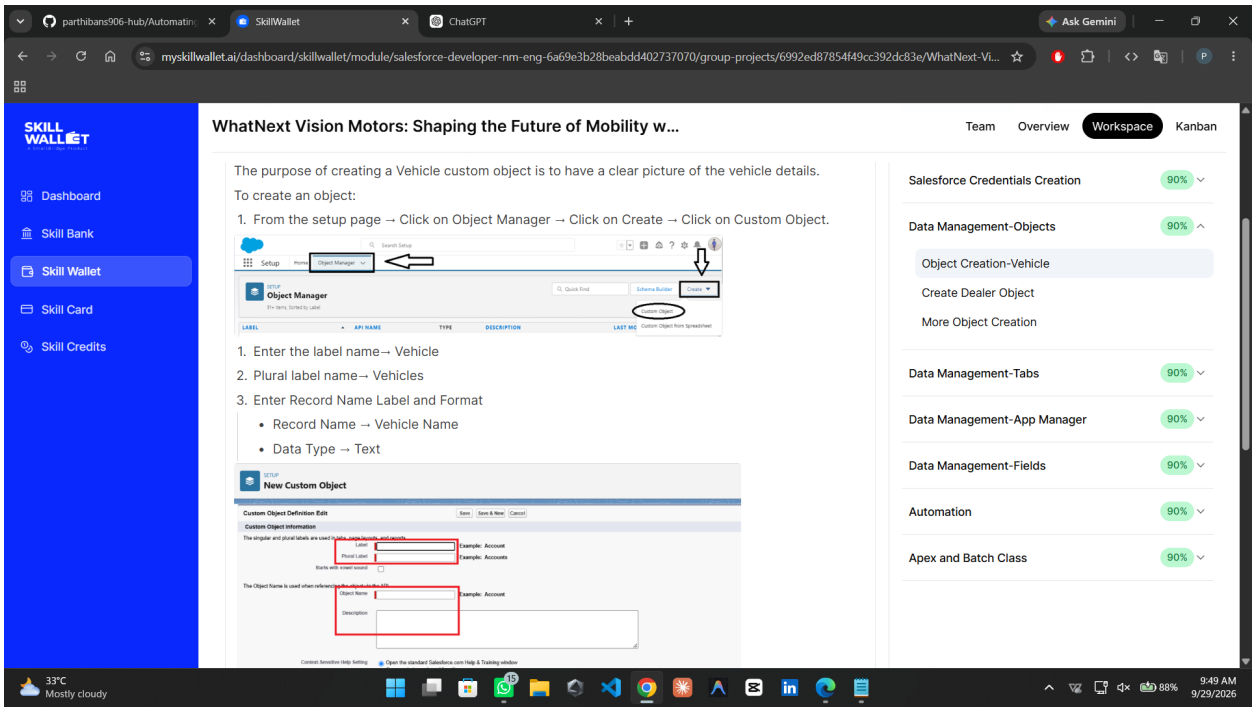


Figure 2. SkillWallet workspace showing custom object creation guidance.

4. Creating the Lightning App

Create a dedicated Lightning App so the project objects and reports can be accessed from one navigation menu.

Navigation: Setup → Quick Find → App Manager → New Lightning App

App details:

Setting	Value
App Name	WhatNext Vision Motors
Developer Name	Auto-populated
Description	Salesforce CRM application for vehicle sales and customer operations
Image	Optional
Primary Color	Keep the default unless branding requires a change

Wizard steps:

- App Details → enter the application information.
- App Options → keep defaults → Next.
- Utility Items → keep defaults → Next.
- Navigation Items → search and add Vehicle, Vehicle Dealer, Customer, Order, Test Drive, Service Request, Reports and Dashboard.
- Select the custom Vehicle Dealer object created in the previous step.
- Add the System Administrator profile under User Profiles as required.
- Finish the app configuration and save.

5. Record-Triggered Flow — Test Drive Reminder

Create a record-triggered flow that sends a customer an email reminder one day before a scheduled test drive.

Step 1: Setup → Flows → New Flow → Record-Triggered Flow → Create.

Step 2 — Trigger:

- Object: **Vehicle Test Drive**.
- Trigger: A record is created or updated.
- Condition Requirements: All Conditions Are Met (AND).
- Field: **Status__c**.
- Operator: Equals.
- Value: **Scheduled**.

Step 3 — Scheduled Path:

- Click **+ Add Scheduled Paths**.
- Label: Reminder Before Test Drive.
- Time Source: **Test_Drive_Date__c**.
- Offset Number: **1**.
- Offset Options: **Days Before**.
- Click Done.

Step 4 — Get Customer Information:

- Add Element → Get Records.
- Label: Get Customer Information.
- Object: **Vehicle_Customer__c**.
- Filter: **Id = {!$Record.Customer__c}**.
- Store only the first matching record.
- Automatically store all fields.

Step 5 — Send Email:

- Add Element → Action.
- Action Type: Send Email.
- Label: Send Test Drive Reminder.
- Subject: **Reminder: Your Test Drive is Tomorrow!**
- Recipient: the customer's Email__c field from the Get Records result.
- Rich-Text-Formatted Body: True.
- Create the email body using a text template/variable.

Step 6: Save the flow with label Test Drive Reminder.

Step 7: Activate the flow. Test it with a Vehicle Test Drive record whose status is Scheduled and whose test-drive date is in the future.

6. Apex Trigger Handler — Stock Validation

The trigger handler separates business logic from the trigger. Before insert/update, it validates stock. After insert/update, it updates stock when an order is confirmed.

Create Apex Class: VehicleOrderTriggerHandler.

7. Vehicle Order Trigger

Create the trigger on the **Vehicle_Order__c** object. The trigger calls the handler for before/after insert and update events.

Why this design?

- **Trigger:** handles Salesforce events only.
- **Handler:** contains reusable business logic.
- **Before event:** stops an invalid/out-of-stock order with addError().
- **After event:** reduces vehicle stock after a confirmed order.

8. Batch Apex — Process Pending Orders

A pending order can remain unconfirmed when the vehicle is unavailable. The batch job periodically checks pending orders and confirms them when stock becomes available.

Create Apex Class: VehicleOrderBatch.

The batch class uses Database.QueryLocator to retrieve pending orders and processes them in manageable batches. A batch size such as 50 can be supplied when the batch is started.

9. Scheduled Apex — Run the Batch Automatically

Create a schedulable class that launches the batch job.

Class Name: VehicleOrderBatchScheduler

Schedule the job from Execute Anonymous in Developer Console or an equivalent Apex execution interface.

Important: The expression 0 0 0 * * ? runs at midnight. The commonly written expression 0 0 12 * * ? runs at 12:00 PM (noon), so use the correct expression for the required schedule.

To verify the schedule, open Setup → Scheduled Jobs and check the job name, next execution time, and status.

Reference screenshot

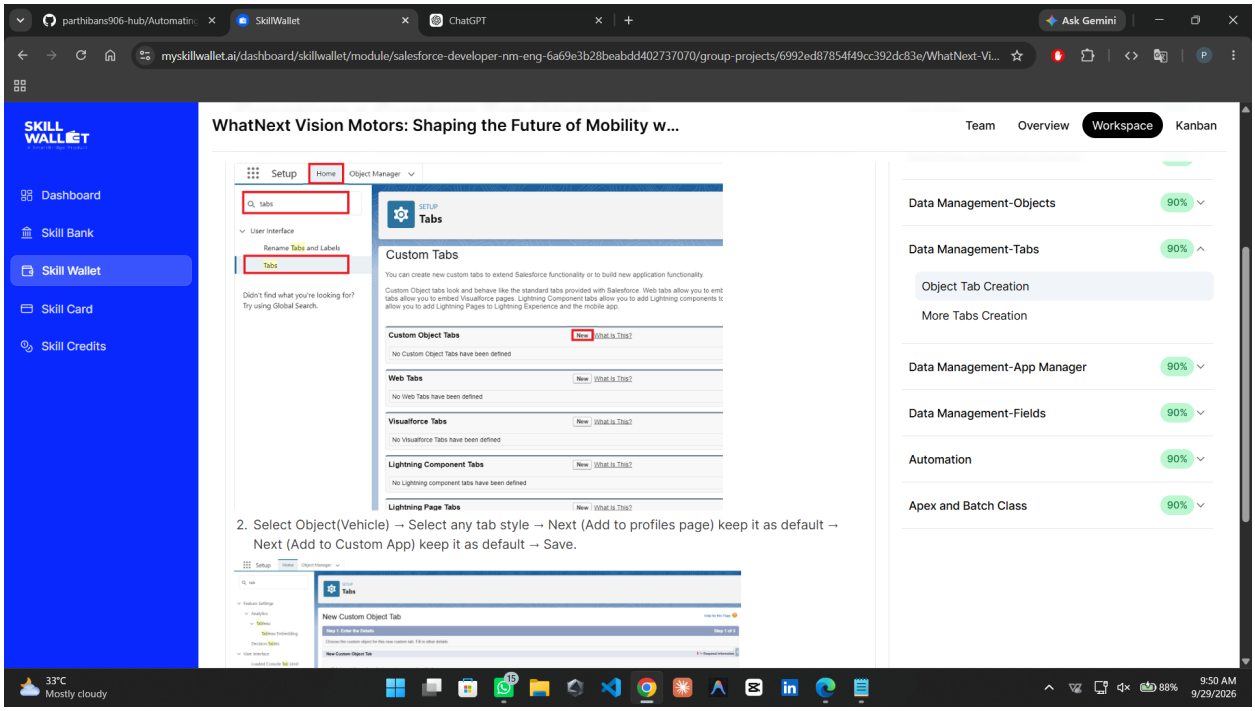


Figure 3. SkillWallet workspace showing the project automation and Apex sections.

10. Testing Checklist

Test Case	Expected Result
Create Vehicle with stock > 0	Vehicle record saves successfully.
Create order for in-stock vehicle	Order can proceed according to configured status/automation.
Create order when stock = 0	Trigger displays: vehicle is out of stock and prevents the order.
Confirm an order	Vehicle stock decreases by 1.
Create Pending order	Order remains Pending while stock is unavailable.
Add new vehicle stock	Scheduled batch can find the pending order.
Run VehicleOrderBatch	Eligible pending order becomes Confirmed and stock decreases.
Create scheduled test drive	Flow schedules the reminder one day before the test drive.
Reach reminder time	Customer receives the configured email reminder.
Check Scheduled Jobs	Daily Vehicle Order Processing appears with its next run.

Recommended validation sequence

- First verify that all custom objects and required fields exist with the exact API names used by Apex.
- Then activate the Flow and test the email path.
- Next test the trigger with both stock available and stock unavailable.
- Finally run the batch manually once, then schedule it and verify the Scheduled Jobs entry.

11. Solution Architecture

Layer	Salesforce Component	Responsibility
Data	Vehicle / Dealer / Customer / Order / Test Drive	Stores CRM records.
UI	Lightning App	Central navigation for project users.
Automation	Record-Triggered Flow	Schedules and sends test-drive reminders.
Validation	Apex Trigger + Handler	Validates stock and updates stock.
Processing	Batch Apex	Processes pending orders in bulk.
Scheduling	Schedulable Apex	Runs the batch automatically.

Business outcome

The completed design creates a connected CRM workflow: customers and vehicles are represented as Salesforce records; order events invoke Apex validation; confirmed orders adjust inventory; pending orders can be processed later through Batch Apex; and scheduled Flow automation keeps customers informed about test drives.

Implementation note

The Apex examples in this model use standard Salesforce custom-field API naming conventions such as Vehicle__c, Status__c and Stock_Quantity__c. Confirm the exact API names in your org before saving code. The supplied source text contained several formatting/escaping issues, so this document presents a cleaned reference version.

WHATSNEXT VISION MOTORS

Salesforce CRM Project — Model Documentation

Completion Summary

- Salesforce Developer Org setup
- Vehicle Dealer custom object
- WhatNext Vision Motors Lightning App
- Test Drive Reminder record-triggered Flow
- Vehicle Order Apex Trigger + Handler
- Vehicle Order Batch Apex
- Scheduled Apex for daily processing
- Testing and verification checklist

End of Model PDF

Use this document as a project record, practical reference, or submission-format model. Replace any placeholder org-specific details and verify the exact field/object API names in your Salesforce Developer Org.